

TokTickIT — Lab 3 Work Log

Issue #34 · Sprint 3 Engineering Contract · branch feature/34-lab3-engineering-contract

1. Overview

This log records the work completed for Lab 3 Issue #34 — writing the Sprint 3 engineering contract required before any Lab 3 implementation branch may start coding, per the handout's spec-before-code rule. This branch changes documentation only; no application code was modified.

Field	Value
Branch	feature/34-lab3-engineering-contract
Base branch	lab3-staging (branched from main)
Issue	#34 — Sprint 3 Engineering Contract
Commit	3d78470 — docs: write Sprint 3 engineering contract (spec, API, UI, test plan)
Files added	docs/lab-03/specification.md, api-spec.md, ui-spec.md, tests.md (4 files, 1,615 lines)
Code changes	None — documentation only

2. What Was Added, by Category

2.1 Specification — docs/lab-03/specification.md

Full Sprint 3 contract in the handout's required 11-section shape: Sprint Goal, Stakeholder Request, Scope (in/out/deliberately-excluded with reasoning), Functional Requirements, Business Rules, UI Specification Summary, Data Changes (new models, changed Ticket fields, status transition matrix, Lab 2 migration plan, seed data), API Contract summary, Acceptance Criteria, Definition of Done, and Assumptions/Decisions.

Item	Count / detail
Functional Requirements	FR-01 – FR-27, across Authentication, Authorization, Requester, IT Staff, Administrator
Business Rules	BR-01 – BR-38 (BR-01–05 are the handout's own given rules)
Acceptance Criteria	AC-01 – AC-28 (AC-01–04 are the handout's own given criteria)
New data models	User, Session, PublicComment, InternalNote
Changed model	Ticket — requesterId retargeted to User; +ownerId, +itPriority, +requesterMarkedResolvedAt/ById; currentStatus becomes an 8-value enum
Migration plan	Expand -> backfill -> contract, with a row-count/FK verification step
Key resolved decision	Administrator has no ticket-queue/detail access in Lab 3 (user management only) — resolves the handout's open point in section 4.3

2.2 API Contract — docs/lab-03/api-spec.md

26 endpoints documented in Lab 2's per-endpoint template (request body / query parameters / responses tables, exact status codes and error bodies), grounded in the actual routes already in server/src/app.ts.

Area	Endpoints
Auth	login, logout, me, change-password (4)
Requester tickets/attachments	Lab 2's 7 existing routes, switched from a requesterId parameter to session identity (7)
Comments / Notes / mark-resolved	comments (GET/POST), notes (GET/POST), mark-resolved (5)

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Area	Endpoints
IT Staff	queue, ticket detail, claim, assign, priority, status (6)
Administrator	list users, create, edit, reset password (4)

2.3 UI Spec — docs/lab-03/ui-spec.md

8 screens/areas specified with layout, states, and responsive behavior at desktop/tablet/mobile widths, reusing Lab 2's Zen Green tokens with a new badge mapping for role/priority/status. Includes a visual-inspection checklist to run against the screenshots before Release Integration.

Screen / area	New or extends
Login	New
Change Password (forced)	New
App Shell (role-scoped nav, logout)	Replaces RequesterBanner / DevRequesterPicker
Requester Ticket Detail — Public Comments + Problem Appears Resolved	Extends Lab 2's TicketDetail
IT Staff Ticket Queue	New
IT Staff Ticket Detail (claim/assign, IT Priority, status, Internal Notes)	New
Administrator User Management	New

2.4 Test Plan — docs/lab-03/tests.md

Written before implementation (Test DD), covering unit, API/integration, UI component, style, responsive, accessibility, security/authorization, migration, regression, and end-to-end categories, with two traceability matrices (AC-> tests, BR-> tests) and exact file paths matching the handout's required repository structure.

Category	Planned rows
Unit (UNIT-xx)	5
API / integration (API-xx)	48
Security / authorization (SEC-xx)	7
UI component (UI-xx)	13
Style (STY-xx)	2
Responsive (RSP-xx)	2
Accessibility (A11Y-xx)	2
Migration (MIG-xx)	4
Regression (REG-xx)	1 (full Lab 1+2 suite)
End-to-end (E2E-xx)	6

Every row's status is honestly recorded as **Planned**, not Pass — this branch is documentation only, and no Lab 3 test file exists in the repository yet. Turning each row Pass is the job of branches #35–#41 as their code lands.

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3. Test Evidence (Actually Run on This Branch)

Since this branch adds no application code, the honest test evidence for it is a regression baseline: Lab 2's existing automated suite, re-run unmodified, to confirm the documentation-only change broke nothing. Output below is copied from the actual run, not projected.

Server suite — `cd server && npm test`

Test file	Tests
tests/lab-02/lookup-lists.api.test.ts	2 passed
tests/lab-02/ticket-detail.api.test.ts	7 passed
tests/lab-02/create-ticket.api.test.ts	12 passed
tests/lab-02/attachments.api.test.ts	8 passed
tests/lab-02/inspect-attachments.api.test.ts	17 passed
tests/lab-02/my-tickets.api.test.ts	29 passed
tests/lab-02/remove-attachment.api.test.ts	16 passed
tests/lab-01/categories.test.ts	1 passed
tests/lab-01/health.test.ts	1 passed

Result: 9 test files, 93/93 tests passed.

Client suite — `cd client && npm test`

Test file	Tests
tests/lab-01/App.test.tsx + api.test.ts	included in 5 files below
tests/lab-02/create-ticket-form.test.tsx	9 passed
tests/lab-02/my-tickets.test.tsx	11 passed
tests/lab-02/ticket-detail.test.tsx	17 passed
(2 remaining lab-01 files)	5 passed

Result: 5 test files, 42/42 tests passed. (stderr lines during this run are expected console.error calls from tests that deliberately exercise error paths — e.g. a rejected upload or a 403 — not failures.)

Combined baseline: 14 test files, 135/135 tests passed, zero regressions from adding the Lab 3 documentation.

Documentation self-checks performed

- Every AC-xx referenced in specification.md §9 has at least one matching row in tests.md §10's traceability matrix.
- Every file path cited in tests.md matches the handout's §12 minimum required repository structure.
- api-spec.md's endpoint list matches specification.md §8's summary table exactly (26 endpoints, both directions checked).
- Every new/changed Prisma field in specification.md §7 is referenced somewhere in api-spec.md's request/response shapes.

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4. Criteria & Scope Compliance

Mapping this branch's output against the handout's grading rubric (Lab_3_sheet.pdf §14) for the parts it feeds directly.

Rubric part	Status after this branch
Part 2 — Spec DD (5 pts): numbered requirements, business rules, permission matrix, acceptance criteria, migration decisions, DoD, evidence the spec predates the main implementation PRs	Satisfied by specification.md; this branch's PR is itself the timestamped evidence that the spec predates every other Lab 3 branch, all of which currently have 0 commits
Part 3 — Test DD (10 pts): planned tests, AC traceability, real file paths, pass results from main	tests.md's plan and traceability matrices are complete; pass results are N/A until #35–#41 land code — tracked as a known gap, not hidden
Part 9 — Zen Green / responsive evidence (5 pts): ui-spec.md with desktop/tablet/mobile screenshots and a visual checklist	ui-spec.md's structure and checklist are complete; screenshots are pending the actual screens (#35–#39) and captured in #40

Definition of Done — items this branch can satisfy now

Done now:

- ✓ docs/lab-03/specification.md, api-spec.md, ui-spec.md, tests.md written and merged to lab3-staging before any other Lab 3 branch's code.
- ✓ Lab 2's full existing test suite (lab-01, lab-02, server + client) still passes unmodified — zero regressions (see §3).
- ✓ No plaintext password or real secret committed in these four files.

Pending later branches (by design — this branch is docs-only):

- Every FR/BR/AC implemented and covered by a passing automated test — pending #35–#41.
- server/tests/lab-03/*, client/.../lab-03/* passing on lab3-staging — pending.
- Migration verified on a DB copy — pending #35 (Authentication Foundation, which owns the migration).
- e2e/lab-03/* passing — pending #40.
- Screenshots under artifacts/lab-03/screenshots/ — pending #35–#40.
- docs/lab-03/reviewer.md, ai-use.md finalized — pending #41 (Release Integration).

5. Lab 3 Branch / Issue Status

Issue	Feature	Branch	Status
#34	Sprint 3 Engineering Contract	feature/34-lab3-engineering-contract	Done — this log
#35	Authentication Foundation	feature/35-authentication-foundation	Not started
#36	Authorization & Requester Regression	feature/36-authorization-requester-regression	Not started
#37	IT Staff Ticket Queue	feature/37-staff-ticket-queue	Not started
#38	IT Staff Ticket Detail & Workflow	feature/38-staff-ticket-detail-workflow	Not started
#39	Administrator User Management	feature/39-admin-user-management	Not started
#40	E2E, Visual & Responsive Evidence	feature/40-e2e-visual-responsive-evidence	Not started
#41	Release Integration	feature/41-release-integration	Not started

6. Next Step

Open a pull request from feature/34-lab3-engineering-contract into lab3-staging for this branch, then start #35 (Authentication Foundation) — the User/Session data model, the Lab 2 -> Lab 3 migration, and the

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login/logout/me/change-password API and screens — which every other remaining branch depends on.